Shi-Ming Wang

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Summary

CS PhD candidate (expected **12/2026**) building deep learning systems in Python and PyTorch, with hands-on experience in graph neural networks, transfer learning across datasets, and large-scale data pipelines on multi-terabyte scientific data. Coursework and side projects span data mining, information retrieval, NLP (BERT/CodeBERT), and brain–computer interfaces.

Education

PhD in Computer Science — September 2023 – Present (Expected Graduation: **12/2026**) National Yang Ming Chiao Tung University (NYCU), Hsinchu, Taiwan **Dissertation:** *Micro-structural and Macro-topological MRI Framework for Predicting Conversion from MCI to Dementia using Graph Neural Networks and Morphological Similarity Networks.* **Research areas:** graph neural networks (GCN / GAT / GraphSAGE), transfer learning, cross-dataset generalization, model interpretability (GNNExplainer), large-scale data pipelines. **Relevant Coursework:** Algorithms (A+), Data Mining, Information Retrieval, Deep Learning, Brain–Computer Interface.

MSc in Medical Imaging and Radiological Science — September 2017 – August 2019 Chang Gung University, Taoyuan, Taiwan

BSc in Biomedical Imaging and Radiological Science — September 2013 – June 2017 China Medical University, Taichung, Taiwan

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Technical Skills

- **Programming Languages:** Python, C, C#, MATLAB
- **Machine Learning / AI:** PyTorch (with CUDA / GPU training), deep learning, graph neural networks (GCN, GAT, GraphSAGE), CNNs, transfer learning, fine-tuning, model interpretability (GNNExplainer)
- **NLP & Information Retrieval:** DistilBERT, CodeBERT, neural ranking, dense retrieval, TF-IDF, BM25
- **Data & Analytics:** NumPy, pandas, scikit-learn, XGBoost, R; statistical modeling, hypothesis testing
- **Pipelines & Infra:** Workflow orchestration (Nipype), shell scripting, batch processing of multi-terabyte datasets on HPC clusters
- **Tools:** Git, Linux/Unix, Bash

Selected Projects (Coursework & Independent)

- **Neural News Recommendation System** — Built a DistilBERT-based news embedding and user-modeling pipeline with a neural ranking network in PyTorch; outperformed an XGBoost baseline (AUC 0.7432).

- **Information Retrieval System for Text–Code Matching** — Implemented TF-IDF and BM25 from scratch, then built a dense-retrieval pipeline using CodeBERT embeddings; benchmarked sparse vs. dense retrieval on a code-search corpus.
- **Deep Learning for EEG Classification** — Trained CNN-based **EEGNet** and **SCCNet** models under subject-independent and fine-tuning schemes; achieved **82.99% accuracy** on a multi-class motor-imagery task.
- **Hybrid EEG Car Control System** — Engineered a real-time hybrid brain–computer interface using alpha-band features and 2-bit control logic to navigate a robotic car; full Python signal-processing + control loop.

Research & Engineering Experience

Brain and Cognitive Sciences Lab, NYCU — May 2020 – Present

Supervisors: Prof. Chih-Mao Huang, Prof. Liwei Chan.

- **Predicting Dementia Conversion via Transfer Learning and Graph Neural Networks (cross-domain application).** Designed and trained GNN models (GCN / GAT / GraphSAGE) in PyTorch with CUDA-accelerated GPU training to classify MCI-to-dementia conversion. Pre-trained on the large public ADNI cohort and fine-tuned on the smaller local ILAS cohort to handle cross-site domain shift; applied **GNNExplainer** for model interpretability.
- **Construction and Topological Analysis of Multiparametric Morphological Similarity Networks.** Built graph-construction pipeline that computes morphological similarity across 100+ brain regions (Schaefer 2018 atlas) per subject from multimodal MRI features, then extracts graph-theoretic features (global efficiency, modularity, small-worldness) for downstream ML.
- **End-to-End Multimodal MRI Pipeline (ILAS cohort).** Developed Python + Nipype pipeline processing **>500 subjects and ~3 TB** of imaging data, including QC, registration, feature extraction, and statistical modeling — runs on HPC with reproducible orchestration.

Computational Imaging Group, UCL Centre for Medical Image Computing — September 2020 – January 2022

Supervisors: Prof. Gary Hui Zhang, Prof. Marco Palombo

- Designed a longitudinal diffusion-MRI analysis pipeline applying advanced microstructural models to subacute/chronic stroke patients; presented as oral session at ISMRM 2022 (London).
- Collaborated remotely with an international research team across UCL, Sorbonne (Paris), and NYCU; used Git and reproducible Python notebooks to share results.

Laboratory of Magnetic Resonance Research, Chang Gung University — August 2017 – August 2019

Supervisor: Prof. Jiun-Jie Wang

- **Deep learning–based diagnosis of Parkinson's disease from diffusion MRI.** Co-developed a CNN-based classifier with a redundancy-removal feature-selection module; published at ISMRM 2019 (Power Pitch).
- Built a reusable longitudinal-analysis pipeline; first-author publication in *NeuroImage: Clinical* (2019).

Selected Publications

1. ***Wang, S.M.***, Wen, H.J., Huang F., Sun C-W., Huang C.M., Wang, S.L. White matter microstructural integrity mediates associations between prenatal endocrine-disrupting chemicals exposure and intelligence in adolescents. **NeuroImage: Clinical**, 45, 103758, Feb. 2025. *(First author.)*
2. Huang, F., Zhou, M., ***Wang, S.M.***, Wu, J., Ali, L., et al. **A deep learning framework with redundancy removal and its diagnostic performance of Parkinson's disease.** ISMRM 2019, Montreal. *(Power Pitch.)*
3. Huang, F., ***Wang, S.M.***, Yan, G., et al. **Learning-based approach for accelerated IVIM imaging in the placenta.** ISMRM 2020. *(Oral scientific session.)*

Languages

- **English:** TOEFL iBT 102 (Nov. 2022) — full professional proficiency.
- **Mandarin Chinese:** Native.